Projects

Project 3.7

Road Mirages

(**Solutions were provided by the project author**.)

1. If $n(y)$ is increasing then
$$\frac{dn}{dy} > 0 \quad \text{and so} \quad \frac{d^2y}{dx^2} > 0.$$
This means the graph of $y(x)$ is concave up.

2. **(a)** The general solution is $y(x) = c_1x + c_2$ which is the equation of a straight line.

(b) The concavity of a straight line is zero.

(c) This solution is physically reasonable since light will travel in a straight line in a uniform medium.

3. **(a)** Now
$$\frac{1}{n}\frac{dn}{dy} = \frac{1}{a}$$
so the differential equation becomes
$$\frac{d^2y}{dx^2} = \left[1 + \left(\frac{dy}{dx}\right)^2\right]\frac{1}{a}.$$

(b) The new differential equation is
$$\frac{du}{dx} = \frac{1+u^2}{a}$$
which is separable. Separating variables we get
$$\frac{du}{1+u^2} = \frac{dx}{a}.$$
After integrating we get
$$\tan^{-1} u = \frac{x+c_1}{a} \quad \text{so} \quad \frac{dy}{dx} = \tan\frac{x+c_1}{a}.$$

(c) This equation has the solution
$$y = -a\ln\cos\left(\frac{x+c_1}{a}\right) + c_2$$
where c_1 and c_2 are arbitrary constants. Choosing new constants x_0 and y_0 the general solution can be written
$$y - y_0 = -a\ln\cos\left(\frac{x-x_0}{a}\right).$$
The graph of $y(x)$ is concave up and has a vertex at the point (x_0, y_0).

(d) Here $x - x_0 = -50$ and $y_0 = 1.19$ so $y = 1.315$ meters. Now find dy/dx:

$$\frac{dy}{dx} = \tan\left(\frac{x - x_0}{a}\right) = 5 \times 10^{-3}$$

so the distance to the shiny spot is $1.315/5 \times 10^{-3} = 263$ meters.

4. (a) Now

$$\frac{1}{n}\frac{dn}{dy} = \frac{1}{2y}$$

so the differential equation becomes

$$\frac{d^2y}{dx^2} = \left[1 + \left(\frac{dy}{dx}\right)^2\right]\frac{1}{2y}$$

which does not have the independent variable x in it. Consequently we make the substitution

$$\frac{dy}{dx} = u \quad \text{and} \quad \frac{d^2y}{dx^2} = u\,\frac{du}{dy}.$$

(b) The new differential equation becomes

$$u\,\frac{du}{dy} = \frac{1 + u^2}{2y}$$

which is also separable. Separating variables we get

$$\frac{u\,du}{1 + u^2} = \frac{dy}{2y}$$

whose solution is $\ln(1 + u^2) = \ln y + \ln c = \ln cy$ for arbitrary $c > 0$.

(c) Exponentiating both sides we get

$$u^2 = cy - 1 \quad \text{or} \quad \pm\frac{dy}{dx} = \sqrt{cy - 1}$$

for $cy - 1 > 0$ or $c > 1$ since $y \geq 1$. This is also separable: we get

$$\pm\frac{dy}{\sqrt{cy - 1}} = dx.$$

After integrating we have

$$\pm\frac{2}{c}\sqrt{cy - 1} = x - d$$

which simplifies to

$$y = \frac{1}{c} + \frac{c}{4}\,(x - d)^2$$

where c and d are arbitrary constants. Let $1/c = y_0$ and $d = x_0$. We then get

$$y - y_0 = \frac{(x - x_0)^2}{4y_0}$$

which is the equation of a parabola opening upward with vertex at (x_0, y_0) for $y_0 \geq 1$.

(d) Here $x - x_0 = -3$ and $y_0 = 1.19$ so $y = 3.08$ meters. Now find dy/dx:

$$\frac{dy}{dx} = \frac{x - x_0}{2y_0} = -\frac{3}{2.38} = -1.261$$

so the distance to the shiny spot is $3.08/1.261 = 2.44$ meters.

Project 3.10

The Ballistic Pendulum

(**Solutions were provided by the project author**.)

1. The auxiliary equation is $m^2 + g/\ell = 0$, so the general solution of the differential equation is

$$\theta(t) = c_1 \cos\sqrt{\frac{g}{\ell}}\,t + c_2 \sin\sqrt{\frac{g}{\ell}}\,t.$$

The initial condtion $\theta(0) = 0$ implies $c_1 = 0$ and $\theta'(0) = \omega_0$ implies $c_2 = \omega_0\sqrt{\ell/g}$. Thus,

$$\theta(t) = \omega_0\sqrt{\frac{\ell}{g}}\,\sin\sqrt{\frac{g}{\ell}}\,t.$$

2. At $\theta_{\max}$, $\sin\sqrt{g/\ell}\,t = 1$, so

$$\theta_{\max} = \omega_0\sqrt{\frac{\ell}{g}} = \frac{m_b}{m_w + m_b}\,\frac{v_b}{\ell}\sqrt{\frac{\ell}{g}} = \frac{m_b}{m_w + m_b}\,\frac{v_b}{\sqrt{\ell g}}$$

and

$$v_b = \frac{m_w + m_b}{m_b}\sqrt{\ell g}\;\theta_{\max}.$$

3. We have $\cos\theta_{\max} = (\ell - h)/\ell = 1 - h/\ell$. Then

$$\cos\theta_{\max} \approx 1 - \frac{1}{2}\,\theta_{\max}^2 = 1 - \frac{h}{\ell}$$

and

$$\theta_{\max}^2 = \frac{2h}{\ell} \qquad \text{or} \qquad \theta_{\max} = \sqrt{\frac{2h}{\ell}}\,.$$

Thus

$$v_b = \frac{m_w + m_b}{m_b}\,\sqrt{\ell g}\,\sqrt{\frac{2h}{\ell}} = \frac{m_w + m_b}{m_b}\,\sqrt{2gh}\,.$$

4. When $m_b = 5\,\text{g}$, $m_w = 1\,\text{kg}$, and $h = 6\,\text{cm}$, we have

$$v_b = \frac{1005}{5}\sqrt{2(980)(6)} \approx 21,797 \text{ cm/s}.$$

Project 8.1

Two-Ports in Electrical Circuits

(**Solutions were provided by the project author**.)

1. Since the terminals are connected directly, $V_1 = V_2$. Voltage drop across the resistance is V_1. The curent through the resistance is $I_1 - I_2$. Thus $V_1 = (I_1 - I_2)R$. We have two equations, $V_2 = V_1 + 0I_1$ and $I_2 = -V_1/R + I_1$. In matrix form

$$\begin{pmatrix} V_2 \\ I_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -1/R & 1 \end{pmatrix} \begin{pmatrix} V_1 \\ I_1 \end{pmatrix}.$$

The transmission matrix is $\begin{pmatrix} 1 & 0 \\ -1/R & 1 \end{pmatrix}$.

2. Current through resistance R_1 is $I_1 - I_2$. Drop in voltage across R_1 is V_1. Ohm's law gives $V_1 = (I_1 - I_2)R_1, I_2 = I_1 - V_1/R_1$. Current through resistance R_2 is I_2. Drop in voltage across R_2 is $V_1 - V_2$. Thus $V_1 - V_2 = I_2R_2, V_2 = V_1 - I_2R_2 = V_1 - (I_1 - V_1R_1)R_2 = V_1(1 + R_2/R_1) - I_1$. Combine $V_2 = V_1(1 + R_2/R_1) - I_1$ and $I_2 = I_1 - V_1/R_1$.

$$\begin{pmatrix} V_1 \\ I_2 \end{pmatrix} = \begin{pmatrix} 1 + R_2/R_1 & -1 \\ -1/R_1 & 1 \end{pmatrix} \begin{pmatrix} V_1 \\ I_1 \end{pmatrix}.$$

The transmission matrix is $\begin{pmatrix} 1 + R_2/R_1 & -1 \\ -1/R & 1 \end{pmatrix}$.

3. Current through resistance R_1 is I_1. Drop in voltage across R_1 is $V_1 - V_2$. Thus $V_1 - V_2 = I_1R_1$ so $V_2 = V_1 - I_1R_1$. Current through resistance R_2 is $I_1 - I_2$. Drop in voltage across R_2 is V_2. Thus $V_2 = (I_1 - I_2)R_2$ and $I_2 = I_1 - V_2/R_2 = I_1 - (V_1 - I_1R_1)/R_2 = -V_1/R_2 + I_1(1 + R_1/R_2)$. Combine $V_2 = V_1 - I_1R_1$ and $I_2 = -V_1/R_2 + I_1(1 + R_1/R_2)$:

$$\begin{pmatrix} V_1 \\ I_2 \end{pmatrix} = \begin{pmatrix} 1 & -R_1 \\ -1/R_2 & 1 + R_1/R_2 \end{pmatrix} \begin{pmatrix} V_1 \\ I_1 \end{pmatrix}.$$

The transmission matrix is $\begin{pmatrix} 1 & -R_1 \\ -1/R_2 & 1 + R_1/R_2 \end{pmatrix}$.

4. **(a)** The transmission matrix is

$$\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix}.$$

(b) Outputs $\begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 6 \end{pmatrix}$; $V = 1$ volt, $I = 6$ amps.

Project 8.2

Traffic Flow

1. Using conservation of flow at each intersection we have

$$\text{A: } x_1 + x_4 = 300 \qquad \text{B: } x_1 + x_2 = 250$$
$$\text{C: } x_2 + x_3 = 100 \qquad \text{D: } x_3 + x_4 = 150.$$

The corresponding system of equations is described by the augmented matrices

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 300 \\ 1 & 1 & 0 & 0 & 250 \\ 0 & 1 & 1 & 0 & 100 \\ 0 & 0 & 1 & 1 & 150 \end{pmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{pmatrix} 1 & 0 & 0 & 1 & 300 \\ 0 & 1 & 0 & -1 & -50 \\ 0 & 0 & 1 & 1 & 150 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Thus

$$\begin{aligned} x_1 &= -x_4 + 300 \\ x_2 &= x_4 - 50 \\ x_3 &= -x_4 + 150. \end{aligned}$$

We see that x_4 must be at least 50 (so that x_2 is nonnegative) and at most 150 (so that x_3 is nonnegative). Taking $x_4 = 100$ we have the solution $x_1 = 200$, $x_2 = 50$, $x_3 = 50$, and $x_4 = 0$. Taking $x_4 = 50$ we have the solution $x_1 = 250$, $x_2 = 0$, $x_3 = 100$, and $x_4 = 50$. The minimum flow along AB corresponds to the smallest possible value of x_1, which occurs when x_4 is the largest it can be, namely, 150. Thus, the minimum value of x_1 is $-150 + 300 = 150$.

2. Using conservation of flow at each intersection and setting the number of vehicles out of each intersection equal to the number into that intersection we have

$$\text{A: } x_1 = 100 + x_4 \qquad \text{B: } 200 + x_2 = x_1$$
$$\text{C: } x_3 = x_2 + 150 \qquad \text{D: } 50 + x_4 = x_3.$$

The corresponding system of equations is described by the augmented matrices

$$\begin{pmatrix} 1 & 0 & 0 & -1 & 100 \\ 1 & -1 & 0 & 0 & 200 \\ 0 & -1 & 1 & 0 & 150 \\ 0 & 0 & 1 & -1 & 50 \end{pmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{pmatrix} 1 & 0 & 0 & -1 & 100 \\ 0 & 1 & 0 & -1 & -100 \\ 0 & 0 & 1 & -1 & 50 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Thus

$$\begin{aligned} x_1 &= x_4 + 100 \\ x_2 &= x_4 - 100 \\ x_3 &= x_4 + 50. \end{aligned}$$

We see that x_4 must be at least 100, in which event x_2 will be 0. When $x_4 = 100$ we have $x_1 = 200$ and $x_3 = 150$.

3. Starting with the right, upper intersection we label the intersections counterclockwise A, B, ... , H. Using conservation of flow at each intersection and setting the number of vehicles out of each intersection equal to the number into that intersection we have

$$\begin{aligned} &\text{A: } x_1 = x_8 + 100 && \text{B: } x_2 + 160 = x_1 \\ &\text{C: } x_3 = x_2 + 80 && \text{D: } x_4 + 100 = x_3 \\ &\text{E: } x_5 = x_4 + 130 && \text{F: } x_6 + 90 = x_5 \\ &\text{G: } x_7 = x_6 + 120 && \text{H: } x_8 + 80 = x_7. \end{aligned}$$

Since each $x_n, n = 1, \ldots, 8$ must be nonnegative, we see from A and B that $x_1 \geq 160$, from C and D that $x_3 \geq 100$, from E and F that $x_5 \geq 130$, and from G and H that $x_7 \geq 120$. Now, if $x_1 = 160$ then $x_2 = 0$ from B, so $x_3 = 80$ from C. But $x_3 \geq 100$ so if $x_3 = 100$ then $x_2 = 20$ from C and $x_1 = 180$. Thus, x_1 is at least 180. When $x_1 = 180$ we find that $x_2 = 20$ (from B), $x_3 = 100$ (from C), $x_4 = 0$ (from D), $x_5 = 130$ (from E), $x_6 = 40$ (from F), $x_7 = 160$ (from G), and $x_8 = 80$ (from H and A). Since these values are all valid, the minimum possible flow along x_1 is 180 vehicles.

4. (a) Using conservation of flow at each intersection and setting the number of vehicles out of each intersection equal to the number into that intersection we have

$$\begin{aligned} &\text{A: } x_1 + x_2 = 200 && \text{B: } x_3 + x_4 = x_1 \\ &\text{D: } 200 = x_4 + x_5 && \text{E: } x_5 = x_2 + x_3. \end{aligned}$$

(b) When $k = 4$,

$$kx_1 + 2kx_2 + kx_3 + 2kx_4 + kx_5 = k(x_1 + 2x_2 + x_3 + 2x_4 + x_5) = 4(x_1 + 2x_2 + x_3 + 2x_4 + x_5).$$

Expressing the system of equations in part **(a)** in reduced echelon form we obtain

$$\begin{aligned} x_1 + x_2 \qquad\qquad\qquad &= 200 \\ x_2 + x_3 + x_4 \qquad &= 200 \\ x_4 + x_5 &= 200. \end{aligned}$$

Adding these equations we see that $x_1 + 2x_2 + x_3 + 2x_4 + x_5 = 600$. Thus, the total time for all 200 vehicles to be in the network is $4 \times 600 = 2400$ minutes or 40 hours. Since there are 200 vehicles, this is an average of 12 minutes per vehicle.

Project 8.15

Temperature Dependence of Resistivity

(**Solutions were provided by the project author**.)

1. $\mathbf{Y} = \begin{pmatrix} 5.6 \\ 5.65 \\ 5.7 \\ 7.82 \\ 11.1 \\ 20.2 \\ 30.5 \end{pmatrix}$ and the matrix $\mathbf{A} = \begin{pmatrix} 0 & 0 & 1 \\ 400 & 20 & 1 \\ 3600 & 60 & 1 \\ 3.24\text{e}4 & 180 & 1 \\ 2.30\text{e}5 & 480 & 1 \\ 4.62\text{e}5 & 680 & 1 \\ 9.60\text{e}5 & 980 & 1 \end{pmatrix}$.

2. The column vector $\mathbf{X}^* = \begin{pmatrix} 2.2\text{e} - 5 \\ 0.00458 \\ 5.57 \end{pmatrix}$

3. Using the values in $\mathbf{X}^*$ above, the resistivity at $300°C$ is

$$5.57 + 0.00458(300 - 20) + (2.2\text{e} - 5)(300 - 20)^2 = 8.55 \text{ ohm-m.}$$

4. The resistivity has increased by a factor $8.55/5.6 = 1.53$, so the resistance becomes $5 \times 1.53 = 7.63$ ohms.

5. The RMS error is 0.90 ohm-m.

6. The RMS error of 0.90 ohm-m roughly means that the least squares quadratic differs from the data by about 1 ohm-m.

7. The predicted resistivity at $2000°C$ is

$$5.57 + 0.00458(2000 - 20) + (2.2\text{e} - 5)(2000 - 20)^2 = 101 \text{ ohm-m.}$$

Extrapolation outside the domain of the data is very unreliable.

Project 9.16

Solutions for Minimal Surfaces

(**Solutions were provided by the project author**.)

1.
$$\begin{aligned} F'(t) &= \iint_R \frac{1}{2\sqrt{1 + \|\nabla u + t\nabla w\|^2}} \frac{d}{dt} \|\nabla u + t\nabla w\|^2 \, dA \\ &= \iint_R \frac{1}{2\sqrt{1 + \|\nabla u + t\nabla w\|^2}} \frac{d}{dt} [(\nabla u + t\nabla w) \cdot (\nabla u + t\nabla w)] \, dA \\ &= \iint_R \frac{1}{2\sqrt{1 + \|\nabla u + t\nabla w\|^2}} \frac{d}{dt} [\|\nabla u\|^2 + 2t\nabla w \cdot \nabla u + t^2\|\nabla w\|^2] \, dA \\ &= \iint_R \frac{1}{2\sqrt{1 + \|\nabla u + t\nabla w\|^2}} [2\nabla w \cdot \nabla u + 2t\|\nabla w\|^2] \, dA \\ F'(0) &= \iint_R \frac{\nabla u}{\sqrt{1 + \|\nabla u\|^2}} \cdot \nabla w \, dA \end{aligned}$$

2. $\displaystyle \oint_c (h\mathbf{F} \cdot \mathbf{n}) \, ds = \iint_R \text{div } (h\mathbf{F}) \, dA = \iint_R (h \, \text{div } \mathbf{F} + (\text{grad } h) \cdot \mathbf{F}) \, dA$

3. Applying the identity from Problem 2 with $\mathbf{F} = \dfrac{\nabla u}{\sqrt{1 + \|\nabla u\|^2}}$ and $h = w$ yields

$$\oint_c w \left(\frac{\nabla u}{\sqrt{1 + \|\nabla u\|^2}} \right) \cdot \mathbf{n} \, ds = \iint_R \left[w \, \text{div} \left(\frac{\nabla u}{\sqrt{1 + \|\nabla u\|^2}} \right) + (\nabla w) \cdot \left(\frac{\nabla u}{\sqrt{1 + \|\nabla u\|^2}} \right) \right] dA$$

but $w = 0$ on C implies $\displaystyle \oint_c w \left(\frac{\nabla u}{\sqrt{1 + \|\nabla u\|^2}} \right) \cdot \mathbf{n} \, ds = 0$ so that

$$F'(0) = \iint_R (\nabla w) \cdot \left(\frac{\nabla u}{\sqrt{1 + \|\nabla u\|^2}} \right) dA = -\iint_R w \, \text{div} \left(\frac{\nabla u}{\sqrt{1 + \|\nabla u\|^2}} \right) dA = 0.$$

4. Using the fact that $||\nabla u||^2 = u_x^2 + u_y^2$ and the Chain Rule we have

$$\frac{\partial}{\partial x}\left(\frac{u_x}{\sqrt{1+||\nabla u||^2}}\right) = \frac{(1+u_x^2+u_y^2)u_{xx} - (u_x^2 u_{xx} + u_x u_y u_{yx})}{(1+u_x^2+u_y^2)^{3/2}}$$

and

$$\frac{\partial}{\partial y}\left(\frac{u_y}{\sqrt{1+||\nabla u||^2}}\right) = \frac{(1+u_x^2+u_y^2)u_{yy} - (u_y u_x u_{xy} + u_y^2 u_{yy})}{(1+u_x^2+u_y^2)^{3/2}}$$

so that

$$\operatorname{div}\left(\frac{\nabla u}{\sqrt{1+||\nabla u||^2}}\right) = \frac{(1+u_y^2)u_{xx} + (1+u_x^2)u_{yy} - 2u_x u_y u_{xy}}{(1+u_x^2+u_y^2)^{3/2}}.$$

5. If u is a function of x only, then $u_y = u_{yy} = 0$ so the minimal surface equation reduces to $u_{xx} = 0$. Antidifferentiating once yields $u_x = f(y)$ where f is an arbitrary function. But since u_x is a function of x only, $u_x = a$ where a is a constant. Antidifferentiating again yields $u(x,y) = ax + b$ where b is a constant. If u is a function of y only then a similar argument yields $u(x,y) = cy + d$ where c and d are constants.

6. If $u = f(r)$ then, since $r = \sqrt{x^2+y^2}$, $u_x = f'(r)\dfrac{\partial r}{\partial x} = f'(r)\dfrac{x}{r}$ and $u_y = f'(r)\dfrac{\partial r}{\partial y} = f'(r)\dfrac{y}{r}$. Then we have

$$\begin{aligned}
u_{xx} &= f''(r)\frac{x^2}{r^2} + f'(r)\left(\frac{1}{r} - \frac{x^2}{r^3}\right)\\
u_{yy} &= f''(r)\frac{y^2}{r^2} + f'(r)\left(\frac{1}{r} - \frac{y^2}{r^3}\right)\\
u_{xy} &= f''(r)\frac{xy}{r^2} + f'(r)\left(\frac{-xy}{r^3}\right).
\end{aligned}$$

Substituting these results into the minimal surface equation yields

$$\left(1+[f'(r)]^2\frac{y^2}{r^2}\right)\left(f''(r)\frac{x^2}{r^2} + f'(r)\frac{y^2}{r^3}\right) + \left(1+[f'(r)]^2\frac{x^2}{r^2}\right)\left(f''(r)\frac{y^2}{r^2} + f'(r)\frac{x^2}{r^3}\right)$$
$$- 2[f'(r)]^2\frac{xy}{r^2}\left(f''(r)\frac{xy}{r^2} - f'(r)\frac{xy}{r^3}\right) = 0.$$

Algebraic rearrangement yields

$$f''(r)\left[1+[f'(r)]^2\left(\frac{x^2y^2}{r^4} + \frac{x^2y^2}{r^4} - \frac{2x^2y^2}{r^4}\right)\right] + f'(r)\left[\frac{1}{r} + [f'(r)]^2\left(\frac{y^4}{r^5} + \frac{x^4}{r^5} + \frac{2x^2y^2}{r^5}\right)\right] = 0,$$

so

$$f''(r) + f'(r)\left(\frac{1}{r} + \frac{[f'(r)]^2}{r}\right) = 0 \quad \text{and} \quad rf''(r) + f'(r)(1+[f'(r)]^2) = 0.$$

7. Letting $g = f'(r)$ we have $r\,\dfrac{dg}{dr} + g(1+g^2) = 0$ and separating variables yields

$$\left(\frac{1}{g} - \frac{g}{1+g^2}\right)dg = -\frac{1}{r}\,dr.$$

Antidifferentiation yields

$$\ln g - \frac{1}{2}\ln(1+g^2) = -\ln r + k, \quad \text{where } k \text{ is a constant,}$$

or

$$\ln\left(\frac{g}{\sqrt{1+g^2}}\right) = \ln\left(\frac{c}{r}\right), \quad \text{where } \ln c = k.$$

Solving for g we obtain

$$f'(r) = g(r) = \frac{c}{\sqrt{r^2 - c^2}} = \frac{1}{\sqrt{r^2/c^2 - 1}} \quad \text{so} \quad u = f(r) = \int \frac{1}{\sqrt{r^2/c^2 - 1}}\, dr.$$

The substitution $r = c\cosh t$ results in $u = ct + d$ where d is a constant, so that finally $r = c\cosh\left(\frac{u-d}{c}\right)$.

8. If $u = f(\theta)$ then, since $\tan\theta = y/x$, $u_x = f'(\theta)(-y/r^2)$ and $u_y = f'(\theta)(x/r^2)$. Then we have

$$\begin{aligned} u_{xx} &= f''(\theta)\frac{y^2}{r^4} + f'(\theta)\left(\frac{2xy}{r^4}\right) \\ u_{yy} &= f''(\theta)\frac{x^2}{r^4} + f'(\theta)\left(\frac{-2xy}{r^4}\right) \\ u_{xy} &= f''(\theta)\left(\frac{-xy}{r^4}\right) + f'(\theta)\left(\frac{2y^2}{r^4} - \frac{1}{r^2}\right). \end{aligned}$$

Substituting these results into the minimal surface equation yields

$$\begin{aligned} &\left(1 + [f'(\theta)]^2\frac{x^2}{r^4}\right)\left[f''(\theta)\frac{y^2}{r^4} + f'(\theta)\left(\frac{2xy}{r^4}\right)\right] + \left(1 + [f'(\theta)]^2\frac{y^2}{r^4}\right)\left[f''(\theta)\frac{x^2}{r^4} + f'(\theta)\left(\frac{-2xy}{r^4}\right)\right] \\ &\qquad + 2[f'(\theta)]^2\frac{xy}{r^4}\left[f''(\theta)\left(\frac{-xy}{r^4}\right) + f'(\theta)\left(\frac{y^2 - x^2}{r^4}\right)\right] = 0. \end{aligned}$$

Algebraic rearrangement yields $\frac{1}{r^2} f''(\theta) = 0$, so $f''(\theta) = 0$ and $f(\theta) = c\theta + d$, where c and d are constants.

Project 14.3

The Hydrogen Atom

(**Solutions were provided by the project author**.)

1. The total mechanical energy for an electron of charge $-e$ and mass m moving in a circular orbit of radius r with velocity v is given by

$$E = \frac{mv^2}{2} + (-e)\frac{e}{4\pi\epsilon_0 r},$$

where we used the fact that the electric potential due to the proton is $V = e/4\pi\epsilon_0 r$. To obtain an expression for the kinetic term mv^2 we use the force balance

$$\frac{mv^2}{r} = \frac{e^2}{4\pi\epsilon_0 r}, \tag{1}$$

where we used the formula for the Coulomb force between the proton and the electron. Inserting this into the expression for the energy gives

$$E = \frac{e^2}{8\pi\epsilon_0 r} - \frac{e^2}{4\pi\epsilon_0 r} = -\frac{e^2}{8\pi\epsilon_0 r}. \tag{2}$$

The classical angular momentum is given by $L = mvr$. We obtain from (**1**) that

$$v = \sqrt{\frac{e^2}{m4\pi\epsilon_0 r}}, \quad \text{which leads to} \quad L = m\sqrt{\frac{e^2}{4\pi\epsilon_0 r}}\, r = \sqrt{\frac{me^2 r}{4\pi\epsilon_0}}.$$

2. Letting

$$n\hbar = \sqrt{\frac{me^2 r}{4\pi\epsilon_0}}$$

leads to the following expression for the orbital radius:

$$r = \frac{4\pi\epsilon_0 n^2\hbar^2}{me^2}\,.$$

Substituting this into (**2**) gives

$$E_n = -\frac{me^4}{(4\pi\epsilon_0)^2 2\hbar^2 n^2}\,. \tag{3}$$

3. The energy difference for a transition from the energy level E_k to the energy level $E_n, k > n$, is

$$E_k - E_n = -\frac{me^4}{(4\pi\epsilon_0)^2 2\hbar^2 k^2} + \frac{me^4}{(4\pi\epsilon_0)^2 2\hbar^2 n^2} = \frac{me^4}{(4\pi\epsilon_0)^2 2\hbar^2}\left(\frac{1}{n^2} - \frac{1}{k^2}\right).$$

Substituting this into Plank's formula $\Delta E = \hbar\nu$ and using the relation $\lambda\nu = c$ gives

$$\frac{1}{\lambda} = \frac{me^4}{8\hbar^3\epsilon_0^2 c}\left(\frac{1}{n^2} - \frac{1}{k^2}\right). \tag{4}$$

If we now set $n = 2$ in the formula above we obtain exactly the Balmer series with $R_H = \dfrac{me^4}{8\hbar^3\epsilon_0^2 c}$.

A literature search provides the following physical constants:

$$\begin{aligned} m &= 9.109534 \cdot 10^{-31}\text{kg} \\ e &= 1.6021892 \cdot 10^{-19}\text{C} \\ \hbar &= 6.626176 \cdot 10^{-34} J \cdot s \\ \epsilon_0 &= 8.854187918 \cdot 10^{-12}\,\frac{\text{C}^2}{\text{N}\cdot\text{m}^2} \\ c &= 2.99792458 \cdot 10^8 m/s. \end{aligned}$$

Inserting these values in the formula for R_H gives

$$R_H = 1.0973731 \cdot 10^7\text{m}^{-1}$$

which is comparable to the experimental value $10967757.6 \pm 1.2\text{m}^{-1}$, but lying outside the error range. To improve the accuracy we can replace the mass m by the reduced mass $mM/(m+M)$, where

$$M = 1.6726485 \cdot 10^{-27}$$

is the mass of the proton. With this correction, we obtain the value

$$R_H = 10967757.78\text{m}^{-1},$$

which is now remarkable close to the best available empirical value.

4. Expressing the Laplacian in spherical coordinates, the Schrödinger equation with a Coulomb potential assumes the form

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2\Psi(\mathbf{x})}{\partial r^2} + \frac{2}{r}\frac{\partial\Psi(\mathbf{x})}{\partial r} + \frac{1}{r^2}\frac{\partial^2\Psi(\mathbf{x})}{\partial\theta^2} + \frac{\cot\theta}{r^2}\frac{\partial\Psi(\mathbf{x})}{\partial\theta} + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\Psi(\mathbf{x})}{\partial\phi^2}\right) - \frac{e^2}{4\pi\epsilon_0 r}\Psi(\mathbf{x}) = E\Psi(\mathbf{x}).$$

If we now substitute the separated solution $\Psi(x) = R(r)\Theta(\theta)\Phi(\phi)$ into this equation, divide each term by Ψ and multiply by r^2 we find

$$-\frac{\hbar^2 r^2}{2m}\left(\frac{R''}{R}+\frac{2}{r}\frac{R'}{R}-\frac{2me^2}{4\pi\epsilon_0 r^3}-\frac{2mE}{\hbar^2 r^2}\right)+\left(\frac{1}{\sin^2\theta}\frac{\Phi''}{\Phi}+\cot\theta\,\frac{\Theta'}{\Theta}+\frac{\Theta''}{\Theta}\right)=0.$$

Since the first term above is a function of r alone while the second depends only on θ and ϕ, they must each be equal to a constant, which we call k. Therefore the radial equation is

$$-\frac{\hbar^2 r^2}{2m}\left(\frac{R''}{R}+\frac{2}{r}\frac{R'}{R}-\frac{2me^2}{4\pi\epsilon_0 r}-\frac{2mE}{\hbar^2}\right)=k,$$

which after some rearrangement of terms becomes

$$R''+\frac{2}{r}R'+\frac{2m}{\hbar^2}\left(\frac{e^2}{4\pi\epsilon_0 r}+E\right)R=-k\,\frac{2m}{\hbar^2 r^2}\,. \qquad (\mathbf{5})$$

5. With $k = 0$ and $r \to \infty$, the radial equation becomes

$$R''+\frac{2mE}{\hbar^2}R=0.$$

Recalling that $E < 0$, we see that the general solution to this equation is a linear combination of the functions e^{Cr} and e^{-Cr} where

$$C=\sqrt{-\frac{2mE}{\hbar^2}}\,.$$

Since the solution cannot explode to infinity as r increases, we discard the term e^{Cr}.

6. If we insert a solution of the form $R(r) = f(r)e^{-Cr}$ into (**5**) (with $k = 0$) we find that f must satisfy

$$f''+\left(\frac{2}{r}-2C\right)f'+\left[C^2-\frac{2C}{r}+\frac{2m}{\hbar}\left(\frac{e^2}{4\pi\epsilon_0 r}+E\right)\right]f=0. \qquad (\mathbf{6})$$

Using the expansion

$$f(r)=a_0+a_1 r+a_2 r^2+\cdots,$$

we see that

$$f'(r)=a_1+2a_2 r+3a_3 r^2+\cdots$$

and

$$f''(r)=2a_2+6a_3 r+\cdots.$$

Inserting these into (**6**) and collecting the coefficients of the terms proportional to $1/r$, 1, r, $r^2, \ldots$ leads to the recurrence relation

$$a_j=2\,\frac{jC-B}{j(j+1)}\,a_{j-1},\quad j=1,2,\ldots,$$

where $B = me^2/4\pi\epsilon_0\hbar^2$ (notice that the coefficient a_0 remains undetermined).

7. We can rewrite the recurrence relation above as

$$a_j=2\,\frac{C-B/j}{j+1}\,a_{j-1},\quad j=1,2,\ldots,$$

which, for large values of j gives

$$a_j=\frac{2C}{j+1}\,a_{j-1}\approx\frac{2C}{j}\,a_{j-1}.$$

But this is the power series for the function e^{2Cr}, which then implies that the function $R(r)$ would behave like $e^{2Cr}e^{-Cr} = e^{Cr}$. Since this is unbounded for large values of r we conclude that the power series for $f(r)$ must terminate at a finite number of terms. It is then easy to see from the original recurrence relation that the only way for this to happen is if $B = nC$ for some integer n.

8. Since

$$B = \frac{me^2}{4\pi\epsilon_0\hbar^2} \quad \text{and} \quad C = \sqrt{-\frac{2mE}{\hbar^2}}\,,$$

the condition $B = nC$ implies that

$$\frac{me^2}{4\pi\epsilon_0\hbar^2} = n\sqrt{-\frac{2mE}{\hbar^2}}$$

or

$$\frac{m^2e^4}{(4\pi\epsilon_0)^2\hbar^4} = -n^2\frac{2mE}{\hbar^2}\,.$$

We conclude that the allowed energies for the hydrogen atom in a state with zero angular momentum are

$$E_n = -\frac{me^4}{(4\pi\epsilon_0)^2\hbar^2n^2}\,.$$

Project 15.4

The Uncertainty Inequality in Signal Processing

(**Solutions were provided by the project author**.)

1. $\hat{f}(-\alpha) = \int_{-\infty}^{\infty} f(x)e^{-i\alpha x}dx = \int_{-\infty}^{\infty} \overline{f(x)e^{i\alpha x}}dx = \overline{\int_{-\infty}^{\infty} f(x)e^{i\alpha x}dx} = \overline{\hat{f}(\alpha)}$

2. Note that

$$\widehat{f_k}(\alpha) = \int_{-\infty}^{\infty} f(x-k)e^{i\alpha x}dx$$

so letting $u = x - k$ yields

$$\widehat{f_k}(\alpha) = \int_{-\infty}^{\infty} f(u)e^{i\alpha(u+k)}du = \int_{-\infty}^{\infty} f(u)e^{i\alpha u}e^{i\alpha k}du = e^{i\alpha k}\int_{-\infty}^{\infty} f(u)e^{i\alpha u}du = e^{i\alpha k}\hat{f}(\alpha).$$

3. Note that

$$\widehat{f_c}(\alpha) = \int_{-\infty}^{\infty} f(cx)e^{i\alpha x}dx$$

so letting $u = cx$ yields

$$\widehat{f_c}(\alpha) = \frac{1}{c}\int_{-\infty}^{\infty} f(u)e^{i\alpha(u/c)}du = \frac{1}{c}\int_{-\infty}^{\infty} f(u)e^{i(\alpha/c)u}du = \frac{1}{c}\hat{f}(\alpha/c).$$

4. Note that

$$\begin{aligned}[D(f_c)\cdot B(f_c)]^2 &= \frac{\int_{-\infty}^{\infty} x^2[f_c(x)]^2dx}{\int_{-\infty}^{\infty}[f_c(x)]^2dx}\cdot\frac{\int_{-\infty}^{\infty}\alpha^2|\widehat{f_c}(\alpha)|^2d\alpha}{\int_{-\infty}^{\infty}|\widehat{f_c}(\alpha)|^2d\alpha} \\ &= \frac{\int_{-\infty}^{\infty} x^2[f(cx)]^2dx}{\int_{-\infty}^{\infty}[f(cx)]^2dx}\cdot\frac{\int_{-\infty}^{\infty}\alpha^2(1/c^2)|\hat{f}(\alpha/c)|^2d\alpha}{\int_{-\infty}^{\infty}(1/c^2)|\hat{f}(\alpha/c)|^2d\alpha}\end{aligned}$$

and with the substitutions $u = cx$ and $\beta = \alpha/c$ this becomes

$$\frac{(1/c^3)\int_{-\infty}^{\infty} u^2[f(u)]^2 du}{(1/c)\int_{-\infty}^{\infty}[f(u)]^2 du} \cdot \frac{c^3\int_{-\infty}^{\infty}\beta^2|\hat{f}(\beta)|^2 d\beta}{c\int_{-\infty}^{\infty}|\hat{f}(\beta)|^2 d\beta} = \frac{\int_{-\infty}^{\infty} u^2[f(u)]^2 du}{\int_{-\infty}^{\infty}[f(u)]^2 du} \cdot \frac{\int_{-\infty}^{\infty}\beta^2|\hat{f}(\beta)|^2 d\beta}{\int_{-\infty}^{\infty}|\hat{f}(\beta)|^2 d\beta}.$$

5. From Example 3 of Section 15.3 in the text it follows that for $f(x) = e^{-|x|}$, $\hat{f}(\alpha) = \dfrac{2}{1+\alpha^2}$. It is easy to check that

$$\int_{-\infty}^{\infty}[f(x)]^2 dx = \int_{-\infty}^{\infty} e^{-2|x|} dx = 2\int_0^{\infty} e^{-2x} dx = 1.$$

Integrating by parts (twice) shows that

$$\int_{-\infty}^{\infty} x^2[f(x)]^2 dx = 2\int_0^{\infty} x^2 e^{-2x} dx = \frac{1}{2}.$$

Finally the substitution $\alpha = \tan\theta$, followed by a partial fraction decomposition, yields

$$\int_{-\infty}^{\infty}[\hat{f}(\alpha)]^2 d\alpha = 4\int_{-\infty}^{\infty}\frac{1}{(1+\alpha^2)^2} d\alpha = 4\int_{-\pi/2}^{\pi/2}\cos^2\theta d\theta = 4(\pi/2) = 2\pi.$$

The fact that

$$\int_{-\infty}^{\infty}\frac{1}{1+\alpha^2}\, d\alpha = \arctan\alpha\Big|_{-\infty}^{\infty} = \pi$$

yields

$$\int_{-\infty}^{\infty}\alpha^2[\hat{f}(\alpha)]^2 d\alpha = 4\int_{-\infty}^{\infty}\frac{\alpha^2}{(1+\alpha)^2}\, d\alpha = 4\int_{-\infty}^{\infty}\frac{1}{1+\alpha^2} - \frac{1}{(1+\alpha^2)^2}\, d\alpha = 4\left(\pi - \frac{\pi}{2}\right) = 2\pi.$$

So finally

$$[D(f)\cdot B(f)]^2 = \frac{\int_{-\infty}^{\infty} x^2[f(x)]^2 dx}{\int_{-\infty}^{\infty}[f(x)]^2 dx} \cdot \frac{\int_{-\infty}^{\infty}\alpha^2|\hat{f}(\alpha)|^2 d\alpha}{\int_{-\infty}^{\infty}|\hat{f}(\alpha)|^2 d\alpha} = \frac{(1/2)}{1}\cdot\frac{2\pi}{2\pi} = \frac{1}{2}.$$

6. (a) By Problem 4 above and (**6**) of Section 15.4 in the text, if $\mathscr{F}\{f(x)\} = F(\alpha)$ and $\mathscr{F}\{g(x)\} = G(\alpha)$ then

$$\int_{-\infty}^{\infty} f(\tau)g(x-\tau)d\tau = \mathscr{F}^{-1}(F(\alpha)G(\alpha)) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\alpha)G(\alpha)e^{-i\alpha x} d\alpha.$$

Taking $g(x) = f(-x)$, and noting that in this case $G(\alpha) = \overline{F(\alpha)}$, it follows that

$$\int_{-\infty}^{\infty} f(\tau)f(\tau - x)d\tau = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\alpha)\overline{F(\alpha)}e^{-i\alpha x} d\alpha.$$

The result follows from taking $x = 0$ in this last equation.

(b) Define a real valued quadratic function by

$$q(\lambda) = \int_a^b [\lambda h_1(s) - h_2(s)]^2\, ds = \left[\int_a^b [h_1(s)]^2 ds\right]\lambda^2 - \left[2\int_a^b h_1(s)h_2(s)ds\right]\lambda + \left[\int_a^b [h_2(s)]^2 ds\right].$$

Since $q(\lambda) \geq 0$ for all λ, the discriminant of $q(\lambda)$ must be less than or equal to 0. So

$$4\left|\int_a^b h_1(s)h_2(s)ds\right|^2 - 4\int_a^b [h_1(s)]^2 ds\int_a^b [h_2(s)]^2 ds \leq 0$$

from which the inequality follows immediately. The case of equality occurs when the discriminant of $q(\lambda)$ is 0, which means that there is exactly one real root c of $q(\lambda)$. For that c, $q(c) = \displaystyle\int_a^b [ch_1(s) - h_2(s)]^2\, ds = 0$ so $ch_1(s) - h_2(s) = 0$ for all s, that is, $h_2 = ch_1$.

(c) By the Schwartz inequality we have:

$$\left|\int_{-\infty}^{\infty} xf(x)f'(x)\,dx\right|^2 \le \left(\int_{-\infty}^{\infty} |xf(x)|^2 dx\right)\left(\int_{-\infty}^{\infty} (f'(x))^2 dx\right).$$

On the left,

$$\int_{-\infty}^{\infty} xf(x)f'(x)\,dx = x\frac{[f(x)]^2}{2}\Big|_{-\infty}^{\infty} - \frac{1}{2}\int_{-\infty}^{\infty}[f(x)]^2 dx = -\frac{1}{2}\int_{-\infty}^{\infty}[f(x)]^2 dx.$$

On the right, by Parseval's formula and operational property (**11**) of Section 15.4 in the text,

$$\int_{-\infty}^{\infty}[f'(x)]^2 dx = \frac{1}{2\pi}\int_{-\infty}^{\infty}\left|\widehat{f'}(\alpha)\right|^2 d\alpha = \frac{1}{2\pi}\int_{-\infty}^{\infty}\alpha^2\left|\hat{f}(\alpha)\right|^2 d\alpha$$

so

$$\frac{1}{4}\left(\int_{-\infty}^{\infty}[f(x)]^2 dx\right)^2 \le \int_{-\infty}^{\infty} x^2[f(x)]^2 dx \cdot \frac{1}{2\pi}\int_{-\infty}^{\infty}\alpha^2\left|\hat{f}(\alpha)\right|^2 d\alpha.$$

Dividing both sides by $\left(\int_{-\infty}^{\infty}[f(x)]^2 dx\right)^2$ we have

$$\frac{1}{4} \le \frac{\int_{-\infty}^{\infty} x^2[f(x)]^2 dx}{\int_{-\infty}^{\infty}[f(x)]^2 dx}\cdot\frac{\int_{-\infty}^{\infty}\alpha^2|\hat{f}(\alpha)|^2 d\alpha}{2\pi\int_{-\infty}^{\infty}[f(x)]^2 dx}$$

and the result follows from one more application of Parseval's formula since

$$2\pi\int_{-\infty}^{\infty}[f(x)]^2 dx = \int_{-\infty}^{\infty}|\hat{f}(\alpha)|^2 d\alpha.$$

7. (a) By the solution of Problem 6(c) above, equality occurs in the uncertainty inequality when equality occurs in the Schwartz inequality used in the first step of the solution. By Problem 6(b), this occurs when $f' = cxf(x)$ for some constant c. Then

$$\begin{aligned}\frac{f'(x)}{f(x)} &= cx\\ \ln|f(x)| &= cx^2/2 + k\\ f(x) &= \pm e^k e^{cx^2/2} = de^{cx^2/2}.\end{aligned}$$

(b) Taking the Fourier transform of both sides of $f' = cxf(x)$ yields

$$\begin{aligned}-i\alpha\hat{f}(\alpha) &= c\widehat{xf(x)}\\ -i\alpha\hat{f}(\alpha) &= c\left(-i\frac{\mathrm{d}}{\mathrm{d}\alpha}\hat{f}(\alpha)\right)\\ \frac{\mathrm{d}}{\mathrm{d}\alpha}\hat{f}(\alpha) &= \frac{1}{c}\alpha\hat{f}(\alpha).\end{aligned}$$

Then proceeding as in part **(a)** we obtain

$$\hat{f}(\alpha) = \hat{f}(0)e^{\alpha^2/(2c)}.$$

Project 15.4

Fraunhofer Diffraction by a Circular Aperture

(**Solutions were provided by the project author**.)

1. $LX + MY = \rho w \cos\theta \cos\psi + \rho w \sin\theta \sin\psi = \rho w \cos(\theta - \psi)$ and $dX\, dY = \rho\, d\rho\, d\theta$

2. Now

$$U(P) = C \int_0^R \int_0^{2\pi} e^{-ik\rho w \cos(\theta-\psi)} \rho\, d\rho\, d\theta.$$

In

$$\frac{i^{-n}}{2\pi} \int_0^{2\pi} e^{ix\cos\alpha} e^{in\alpha} d\alpha = J_n(x)$$

We choose $n = 0$ and $x = -k\rho w$ and $\alpha = \theta - \psi$ and $d\alpha = d\theta$ to get

$$J_0(-k\rho w) = J_0(k\rho w) = \frac{1}{2\pi} \int_0^{2\pi} e^{-ik\rho w \cos(\theta-\psi)} d\theta$$

for any ψ so we choose $\psi = 0$. Then

$$U(P) = 2\pi C \int_0^R J_0(k\rho w) \rho\, d\rho$$

as required.

3. Using

$$\frac{d}{du}\left[u^{n+1} J_{n+1}(u)\right] = u^{n+1} J_n(u)$$

with $n = 0$ we integrate both sides to get

$$\int_0^x u J_0(u)\, du = x J_1(x) - 0 J_1(0) = x J_1(x).$$

4. So let $u = k\rho w$ and since

$$U(P) = 2\pi C \int_0^R J_0(k\rho w) \rho\, d\rho$$

we get

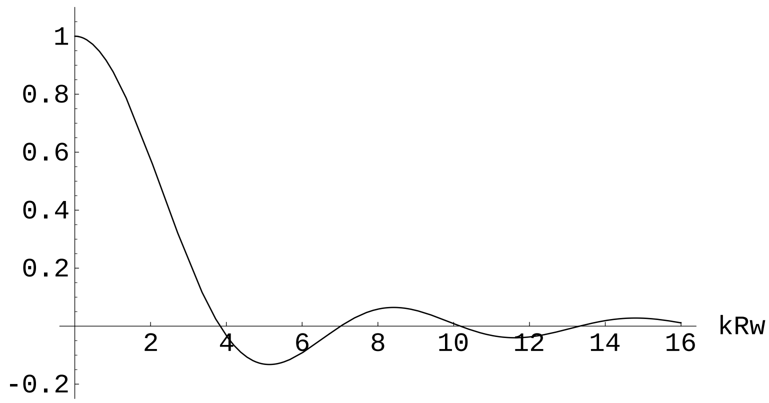
$$\begin{aligned} U(P) &= 2\pi C \int_0^{kwR} \frac{J_0(k\rho w) \rho k w\, d(\rho k w)}{(kw)^2} \qquad \text{(with } kw = \text{constant)} \\ &= 2\pi C \frac{kRw J_1(kRw)}{(kw)^2} = C\pi R^2 \frac{2J_1(kRw)}{kRw}. \end{aligned}$$

5. 1 since $J_1(kRw)$ behaves like $kRw/2$ for small kRw.

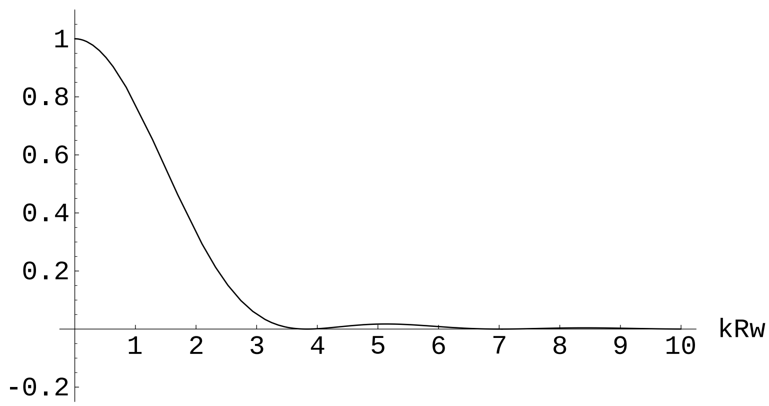
6. I_0 is the maximum intensity in the diffraction pattern and occurs at the location of the geometric image of the star at point O.

7. The smallest nonzero root of $J_1(u)$ is $u = 3.832$ approximately. So the angular radius of the central diffraction disk is $w = 3.832/kR = 0.69$ arc seconds.

8. Here is the amplitude.



and below is the intensity which is the amplitude squared.



9. If the radius R is doubled, the angular width of the diffraction pattern is halved.

10. If the wavelength is doubled, the angular width of the diffraction pattern is doubled.

11. If the focal length of the lens is doubled, then the angular width stays the same.

12. $U(P) = 2\pi C \displaystyle\int_{kwa}^{kwb} \frac{J_0(k\rho w)\rho kw\, d(\rho kw)}{(kw)^2} = C\pi \left(b^2 \frac{2J_1(kbw)}{kbw} - a^2 \frac{2J_1(kaw)}{kaw} \right)$

13. Use

$$U(P) = C\pi \left(b^2 \frac{2J_1(kbw)}{kbw} - a^2 \frac{2J_1(kaw)}{kaw} \right).$$

Let $b = a + \Delta a$. With some rearranging we get

$$U(P) = C\,\frac{2\pi\Delta a}{kw} \left[\frac{kw(a+\Delta a)J_1(kw(a+\Delta a)) - kwaJ_1(kwa)}{kw\Delta a} \right].$$

Notice the quantity in brackets is approximately

$$\frac{d(uJ_1(u))}{du} = uJ_0(u)$$

with $u = kwa$ so $U(P) = C(2\pi a\Delta a)J_0(kwa)$ as required.

Project 16.2

Instabilities of Numerical Methods

(**Solutions were provided by the project author**.)

1. Substitution of $u_{ij} = a_{l,j}\sin(\kappa_l i)$ into the method (**7**) results in the equation

$$[\alpha - 2\cos(\kappa_l)]a_{l,j+1} = [2\cos(\kappa_l) - \beta]a_{l,j}.$$

The factor Q_l in the one-step iteration scheme (**4**) is

$$Q_l = \frac{1-\lambda(1-\cos\kappa_l)}{1+\lambda(1-\cos\kappa_l)},$$

where we have used $\alpha = 2(1+1/\lambda)$ and $\beta = 2(1-1/\lambda)$. Since $\lambda > 0$ and $1-\cos\kappa_l > 0$ for $l = 1, 2, \ldots, n$, we have $-1 < Q_l < 1$ for $l = 1, 2, \ldots, n$, such that the stability constraint (**5**) is met.

2. Substitution of $u_{ij} = a_{l,j}\sin(\kappa_l i)$ into the method (**8**) results in the two-step iteration scheme (**9**). Let $a_{l,j+1} = Q_l a_{l,j}$ and $a_{l,j} = Q_l a_{l,j-1}$. The factor Q_l then satisfies the quadratic equation

$$Q_l^2 + 4\lambda(1-\cos\kappa_l)Q_l - 1 = 0.$$

The quadratic equation has two roots:

$$Q_l^{\pm} = -2\lambda(1-\cos\kappa_l) \pm \sqrt{4\lambda^2(1-\cos\kappa_l)^2 + 1}\,.$$

Since $\lambda > 0$ and $1-\cos\kappa_l > 0$ for $l = 1, 2, \ldots, n$, we have $Q_l^{\pm} > 1$ for any $l = 1, 2, \ldots, n$, such that the stability constraint (**5**) is violated for any $\lambda > 0$.

3. Substitution of $u_{ij} = a_{l,j}\sin(\kappa_l i)$ into the method (**10**) results in the two-step iteration scheme

$$a_{l,j+1} = 2(1-\lambda^2+\lambda^2\cos\kappa_l)a_{l,j} - a_{l,j-1}.$$

By using the same method as in Problem 2, we obtain the quadratic equation for the factor Q_l:

$$Q_l^2 - 2(1-\lambda^2+\lambda^2\cos\kappa_l)Q_l + 1 = 0.$$

The quadratic equation has two roots:

$$Q_l^{\pm} = 1-\lambda^2+\lambda^2\cos\kappa_l \pm \sqrt{(1-\lambda^2+\lambda^2\cos\kappa_l)^2 - 1}\,.$$

It is clear from the quadratic equation that $Q_l^+Q_l^- = 1$. If the two roots are complex, then they are complex conjugates and $Q_l^+Q_l^- = |Q_l^+|^2 = |Q_l^-|^2 = 1$, such that the stability constraint (**5**) is met. If the two roots are distinct and real, then either $|Q_l^+| > 1, |Q_l^-| < 1$ or $|Q_l^+| < 1, |Q_l^-| > 1$, such that the stability constraint (**5**) is violated. The case of distinct and real roots occur when the discriminant of the quadratic equation is positive, i.e.

$$(1-\lambda^2+\lambda^2\cos\kappa_l)^2 - 1 = 4\lambda^2\sin^2\left(\frac{\pi l}{2n}\right)\left[\lambda^2\sin^2\left(\frac{\pi l}{2n}\right) - 1\right] > 0.$$

When $\lambda^2 \le 1$, the above inequality is never true. When $\lambda^2 > 1$, the above inequality is true for $l = n$.

4. Substitution of $u_{i,j} = a_{l,j}e^{\mathrm{i}\kappa_l i}$ into the method (**11**) results in the one-step iteration scheme (**4**) with the complex-valued factor Q_l:

$$Q_l = 1 - \lambda + \lambda e^{\mathrm{i}\kappa_l} = 1 - \lambda + \lambda \cos\kappa_l - \mathrm{i}\lambda \sin\kappa_l\,.$$

The squared absolute value of Q_l is

$$|Q_l|^2 = 1 - 2\lambda(1-\lambda)(1-\cos\kappa_l).$$

Since $\lambda > 0$ and $1 - \cos\kappa_l > 0$ for $l = 1, 2, \ldots, n$, we have $|Q_l| \leq 1$ for $\lambda \leq 1$, such that the stability constraint (**5**) is met. If $\lambda > 1$, we have $|Q_l| > 1$ at least for $l = n$.

5. Substitution of $u_{i,j} = a_{l,j}e^{\mathrm{i}\kappa_l i}$ into the method (**12**) results in the one-step iteration scheme (**4**) with the complex-valued factor Q_l:

$$Q_l = \frac{2}{2 + \lambda e^{\mathrm{i}\kappa_l} - \lambda e^{-\mathrm{i}\kappa_l}} = \frac{1}{1 + \mathrm{i}\lambda \sin\kappa_l}\,,$$

such that

$$|Q_l|^2 = \frac{1}{1 + \lambda^2 \sin^2\kappa_l} < 1.$$

Therefore, the stability constraint (**5**) is met for any $l = 1, 2, \ldots, n$.